

Robust, Fire-Safe, Monomer-Recovery, Highly Malleable Thermosets from Renewable Bioresources

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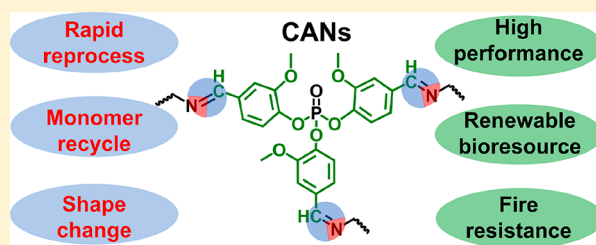
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Supporting Information

ABSTRACT: Conventional thermosets are built by nonrenewable fossil resources and are arduous to be reprocessed, recycled, and reshaped due to their permanent covalent cross-linking, and their flammability makes them unsafe during use. Here, for the first time, we synthesized a novel Schiff base precursor from abundant and renewable lignin derivative vanillin and produced malleable thermosets (Schiff base covalent adaptable networks (CANs)) combining high performance, super-rapid reprocessability, excellent monomer recovery, and arbitrary permanent shape changeability as well as outstanding fire resistance. The Schiff base CANs exhibited high glass transition temperatures of ~ 178 °C, tensile strength of ~ 69 MPa, tensile modulus of ~ 1925 MPa, excellent flame retardancy with UL-94 V0 rating and V1 rating, and high LOI of $\sim 30\%$. Meanwhile, three Schiff base CANs showed high malleability with the activation energy of the bond exchange of $49\text{--}81$ kJ mol^{−1} and could be reprocessed in 2–10 min at 180 °C. These Schiff base CANs provide a prime example to foster the development of advanced thermosetting materials from renewable bioresources.



1. INTRODUCTION

Biomass or bioresource is abundant, inexpensive, and more significantly renewable. In the past two decades, numerous efforts have been made to produce alternatives from biomass to fuels, chemicals, or materials from limited but constantly and rapidly depleting fossil resources which generated climate change and other environmental issues. Thermosets with an estimated annual production of ~ 65000000 tons, depending on fossil resources, have played an important and irreplaceable role in electronic packaging materials, adhesives, composites, coatings, etc., owing to the excellent dimensional stability and outstanding mechanical and thermal properties from their covalently cross-linked network structures.¹ In recent years, there are plenty of thermosets reported from bioresources such as plant oil, rosin, lignin, itaconic acid, isosorbide, cardanol, vanillin, etc.^{2,3} Among these bioresources, vanillin is currently the only bio-based aromatic monomer that is industrialized.⁴ Many types of polymers have been prepared by vanillin, especially the high performance thermosets such as epoxy resin⁵ or polybenzoxazines.⁶ However, these researches mainly focused on the renewability of resources, seldom resolved thermosets, other urgent issues such as recycle trouble, flammability, etc.^{5,7–9}

Because of the chemically cross-linked networks, thermosets are difficult to be recycled or decomposed after curing;^{10,11} the present treatments are incineration and landfill which will give rise to considerable environmental pollution and tremendous waste of resources. In recent years, vitrimers or covalent

adaptable networks (CANs) based on the reversible Diels–Alder reaction,^{12–14} disulfides,^{15–17} —Si—O—Si— exchange,^{18,19} transalkylation,^{20,21} transesterification,^{10,22–24} transcarbamoylation,^{25,26} Schiff base,^{27,28} dioxaborolane metathesis,^{29–31} olefin metathesis,³² etc., attracted great attention to achieve the reprocess or recycle of cross-linked polymeric materials, especially after the seminal vitrimer based on transesterification was reported by French scientist Ludwik Leibler and co-workers.¹⁰ Vitrimers or CANs behave like permanently cross-linked thermosetting materials at service temperatures and are insoluble but can still flow when heated which endowed the malleability corresponding to the reprocessability, thermal healing, etc.^{29,30} Although important advances were achieved, design malleability thermosets combining high performance, readily reprocessability, and monomers recovery is still a challenge.³⁰ The vast majority of vitrimers or CANs exhibited poor thermal and mechanical properties and required high temperature and long time to be reprocessed, and most of these reported CANs or vitrimers needed specific catalyst or high energy input to activate the exchange process, which might arise the risk of catalyst instability and thermal degradation of matrix. Thermosets with relatively high performance based on transesterification³³ and Si—O—C exchange³⁴ often combined long reprocess times of more than 6 h at high temperature to get

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recycled materials with high maintained performance, and phenyl disulfide-containing thermosets³⁵ depended on much expensive phenyl disulfide compounds. Schiff base is the one of the most often employed reversible covalent interaction including three distinct processes of imine condensation/hydrolysis, imine exchange, and imine metathesis.^{36,37} The wide variety of commercially available diamines and dialdehydes makes Schiff base resins or polyimines with a multitude of well-demonstrated unique functionalities highly accessible.²⁸ In 2004, Lehn and co-worker³⁸ reported a linear reversible covalent polyacylhydrazones which were described as dynamers and presented constitutional dynamic diversity due to the imine groups. Until now there have been some reports of polyimines as superior recyclable, degradable, malleable polymeric materials without catalyst, and the reprocess time was much shorter than other CANs or vitrimers.^{27,28,39–44} However, the reported reprocessable Schiff base thermosets also displayed unsatisfactory thermal and mechanical properties with T_g of ~ 61 °C and tensile strength of ~ 42 MPa. Linear vanillin-based Schiff base polymers and their chelation with metal ions were also reported, while their properties were not reported.⁴⁵

The flammability of thermosets blocks their application in the fire resistance-required fields,⁵ so flame retardants are often used together with thermosets to meet the safety requirement. Phosphorus-containing compounds have been regarded as effective and environment friendly flame retardants for polymers and have attracted intensive interests⁴⁶ after some halogenated fire retardants were banned by the European Union attributed to their toxicity. Combined with the issue of fossil resources, the phosphorylation of renewable resources attracts increasing attention in recent years. Some phosphorus compounds from renewable sources were reported, especially for the biophenols with excellent char formation ability. These biophenols, including macromolecular biophenols (lignins and lignocellulosic materials) and some bio-based phenolic compounds such as phenol, resorcinol, phloroglucinol, ferulic alcohol, and vanillin, were used to prepare phosphorus compounds or polymers for flame retardants.⁴⁷ Among these biophenols, renewable lignin derivative vanillin has already showed great potential to produce bio-based phosphorus-containing thermosets with high performance.⁵

Thus, for the first time, the recycle and flammability issues of the thermosets were addressed simultaneously, and vanillin was used as the raw materials. A novel trialdehyde monomer TFMP was synthesized from phosphorus oxychloride and vanillin according to the synthetic route of triphenylphosphate from phosphorus oxychloride and eugenol⁴⁸ and then cured with three different diamines and TFMP through Schiff base reaction to achieve Schiff base CANs. The obtained Schiff base CANs could be readily recycled by two methods of reprocessing with short time and degradation to original monomer TFMP under mild condition and exhibited excellent flame retardancy as high as UL-94 V0 rate during vertical burning. Besides, the Schiff base CANs exhibited high T_g s and mechanical properties, and arbitrary change of steady-state shape was achieved above T_g because of the dynamic covalent Schiff base structure.

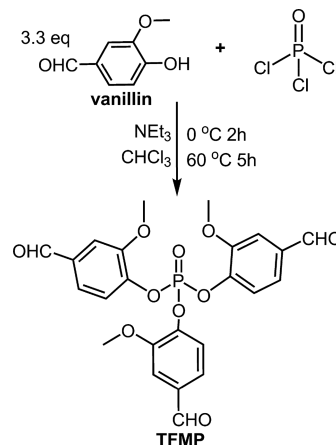
2. EXPERIMENTAL SECTION

2.1. Materials. Vanillin, 4,4-diaminodiphenylmethane (MDA), 4,4-diaminodicyclohexylmethane (PACM), hexamethylenediamine (HMD), and triethylamine were purchased from Aladdin-Reagent Co., China. Phosphorus oxychloride, trichloromethane, ethanol, tetrahydrofuran (THF), and petroleum ether (boiling range: 60–90

°C) were obtained from Sinopharm Chemical Reagent Co., Ltd., China.

2.2. Synthesis of Tris(4-formyl-2-methoxyphenyl) Phosphate (TFMP). TFMP was synthesized by a quintessential condensation reaction between vanillin and phosphorus oxychloride (Scheme 1). 100.42 g (0.66 mol) of vanillin and 66.79 g (0.66 mol) of

Scheme 1. Synthetic Route of TFMP



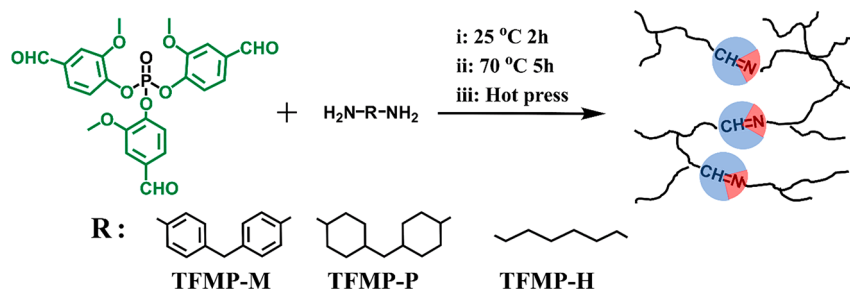
triethylamine (the deacid reagent) were dissolved in 200 mL of trichloromethane in a 500 mL three-neck flask equipped with a mechanical stirrer. 30.67 g (0.2 mol) of phosphorus oxychloride dissolved in 50 mL of chloromethane was added dropwise into the flask over a period of 0.5 h at 0 °C, and they were reacted at 0 °C for 2 h and 50 °C for 6 h. Finally, the mixture was poured into 1000 mL of petroleum ether and stirred for 0.5 h after being cooled to room temperature. A yellow solid was collected by filtration; then the white powder product TFMP (79.6 g) with the yield of around 79.4% was obtained after being washed with ethanol twice followed by drying at 70 °C for 10 h in a vacuum oven. TOF-MS (m/z): 501.0816; mp (°C): 151.2. FTIR (cm^{-1}): 1700 (C=O), 1280 (P=O), 1025 (P–O). ^1H NMR (CDCl_3 , ppm): δ = 3.86 (s, 9H), 7.49 (dd, 6H), 7.62 (d, 3H), 9.97 (s, 3H). ^{13}C NMR (CDCl_3 , ppm): δ = 56.12 (s, 3C), 111.18 (s, 3C), 121.66 (d, 3C), 124.62 (s, 3C), 134.64 (s, 3C), 144.3 (d, 3C), 151.32 (d, 3C), 190.78 (s, 3C). ^{31}P NMR (CDCl_3 , ppm): δ = -18.63 (s, 1P).

2.3. Preparation of the Schiff Base CANs. TFMP was cured by MDA, PACM, and HMD (Scheme 2). TFMP mixed with MDA (or PACM or HMD) in an equivalent ratio of 2:3 was dissolved in trichloromethane (10 times of the total weight of TFMP and MDA (or PACM or HMD)), heated at 50 °C for 30 s, and then added to several 10 cm $\times$ 10 cm $\times$ 5 mm poly(tetrafluoroethylene) molds. The mixture was polymerized and evaporated in a fume hood at room temperature for 5 h and at 70 °C in a blow oven for 2 h to get the precured Schiff base CANs (Figure S1) followed by being pressed with a plate vulcanizer at 180 °C for 3 min (1 min for removing bubbles, 2 min for heat pressing) to get films (Figure S1) with thickness of ~ 300 μm for thermal, mechanical, reprocess, and degradation properties examination. The precured Schiff base CANs were also put into stainless steel molds and platen at 180 °C for 10 min to obtain samples with dimensions of 80 mm $\times$ 6.5 mm $\times$ 3.2 mm for the limit oxygen index (LOI) tests and 130 mm $\times$ 13 mm $\times$ 3.2 mm for the UL-94 vertical burning tests. Both films and rectangular samples were postcured at 180 °C for 2 h in a vacuum oven.

2.4. Gel Content Test. Three Schiff base CANs (around 500 mg) were separately put into the Soxhlet extractor for extracting with tetrahydrofuran for 72 h and then dried in a ventilated oven at 70 °C for 12 h. m_0 is initial mass, and m_1 is final mass after drying; the gel content is calculated by m_1/m_0 .

2.5. Reprocessing Recycle. The reprocess recycling test was performed in a plate vulcanizer. The films were cut into small pieces and placed between two steel sheets covered with two polyimine films (to

Scheme 2. Preparation of Schiff Base CANs



prevent Schiff base CANs adhering on the steel sheets) and hot pressed at 180 °C under a pressure of 15 MPa for 10 min (TFMP-M), for 8 min (TFMP-P), or for 2 min (TFMP-H). After cooling to room temperature, recycled films were obtained. For the second physical recycling, TFMP-M, TFMP-P, and TFMP-H samples were hot pressed at 180 °C under a pressure of 15 MPa for 20, 17, and 3 min, respectively.

2.6. Degradation Recycle and Monomer Recovery. 200 mg of the TMFP-M (or TMFP-P or TMFP-H) was placed in 10 mL of a solution of THF, HCl, and deionized water in a 25 mL vial. The degradation behavior of Schiff base CANs was investigated by adjusting the polymer structures, temperature, HCl concentration, and solvent ratio, and the data are recorded in Table 2. 3 g of the TMFP-M (or TMFP-P or TMFP-H) was placed in 30 mL of a solution of THF and 1 M HCl aqueous solution (8:2, v:v) for monomer recovery. After keeping at 20 °C 24 h, the solution was poured into deionized water (1:5, v:v) and magnetically stirred for 10 min; the solid powder was precipitated, then washed twice with ethanol, and dried under vacuum at 70 °C for 5 h. The obtained yellow solid powder was TFMP, and its recovery rates from TFMP-M, TFMP-P, and TFMP-H were 76.6%, 69.3%, and 73.5%, respectively.

2.7. Arbitrary Steady-State Shape Change. TFMP-H was chosen as an example to investigate the shape changeability of the Schiff base CANs. TFMP-H was bent into different shapes in silicon oil at 100 °C and kept at 100 °C for 2 min and then cooled to room temperature in the air; then TFMP-H samples with different shapes were put in a 100 °C oven for 2 h, and the photos of the original samples and the reshaped samples before and after heating in the 100 °C oven for 2 h were recorded.

2.8. Characterization Methods. ^1H NMR, ^{13}C NMR, and ^{31}P NMR spectra were measured with an AVANCE III Bruker NMR spectrometer (Bruker, Switzerland) with CDCl_3 as solvent. The TOF-MS spectrum was measured by a TripleTOF 4600 time-of-flight mass spectrometer (AB Sciex, America). The FTIR spectra were recorded by a Micro-FTIR Cary660 (Agilent, America) using the powder and films, and the absorbance mode was used. The temperature-dependent Fourier transform infrared measurements were measured with a NICOLET 6700 (Thermo, America) using the KBr pellet method from 50 to 200 °C. Differential scanning calorimetry (DSC) measurements were performed on a Mettler-Toledo TGA/DSC1 thermogravimetric analyzer (METTLER TOLEDO, Switzerland) under a nitrogen atmosphere for glass transition temperature (T_g) with a heating rate of 20 °C min^{-1} and for melting point of TFMP with a heating rate of 10 °C min^{-1} . Thermogravimetric analysis (TGA) was carried using a Mettler-Toledo TGA/DSC1 thermogravimetric analyzer (METTLER TOLEDO, Switzerland) from 50 to 700 °C with a heating rate of 10 °C min^{-1} under a nitrogen and air atmosphere, respectively. The Schiff base CANs with dimensions of 35 mm $\times$ 0.35 mm $\times$ 300 μm were used to examine tensile properties by a Universal Mechanical Testing Machine (Instron 5569A, America) with a cross-head speed of 2 mm min^{-1} . The tensile properties of each sample were recorded as the average of five measurements. Stress relaxation of the Schiff base CANs was tested on a Q800 DMA (TA Instruments, America). The samples with dimensions of 20 mm $\times$ 0.35 mm $\times$ 300 μm were initially preloaded by 1×10^{-3} N force to maintain straightness. After reaching the testing temperature, it was allowed 10 min to reach thermal equilibrium. The specimen was stretched by 2% on the DMA machine,

and the deformation was maintained throughout the test. The decrease of the stress relaxation modulus was recorded. The limit oxygen index (LOI) was conducted according to ASTM D2863-10 on a JF-3 oxygen index instrument (JiangningAnalysis Instrument Company, China) with sample dimensions of 80 mm $\times$ 6.5 mm $\times$ 3.2 mm. UL-94 vertical burning tests were obtained on an AG5100B vertical burning tester (ZhuhaiAngui Testing Equipment Company, China) with sample dimensions of 130 mm $\times$ 13 mm $\times$ 3.2 mm according to ASTM D2863-97. Scanning electron microscopy (SEM, EVO18) was used to observe the morphologies of residues collected after LOI tests with an accelerating potential of 20 kV. The X-ray photoelectron spectroscopy (XPS) spectra was performed on an AXIS ULTRA apparatus (Kratos, England) with the char residues (surface and inside (cut off the surface layer with thickness of 2 mm)).

3. RESULTS AND DISCUSSION

3.1. Synthesis and Characterization of TFMP. TFMP was synthesized by condensation reaction between vanillin and

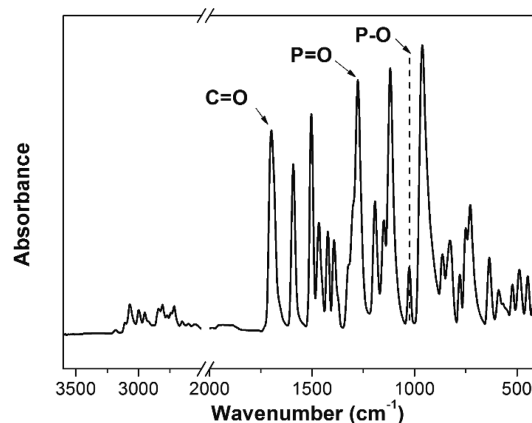


Figure 1. FTIR spectrum of TFMP.

phosphorus oxychloride, and its chemical structure was confirmed by FTIR, ^1H NMR, ^{13}C NMR, ^{31}P NMR, and TOF-MS. In the FTIR spectrum of TFMP (Figure 1), there are characteristic peaks at 1700 cm^{-1} belonging to $\text{C}=\text{O}$, peaks at around 1280 cm^{-1} ascribed to $\text{P}=\text{O}$, and peaks at around 1025 cm^{-1} representing $\text{P}-\text{O}$. In the ^1H NMR of TFMP (Figure 2a), the peak at 3.86 ppm belongs to the H atom of $-\text{CH}_3$, the peaks at 7.45–7.65 ppm represent the H atom of benzene ring, and the peak at 9.97 ppm corresponds to the H atom of $-\text{CHO}$. In the ^{13}C NMR of TFMP (Figure 2b), the peak at 56.12 ppm belongs to the C atom of $-\text{CH}_3$, the peaks at 111.18, 121.66, 124.62, 134.64, 144.3, and 151.32 ppm are ascribed to the C atoms of benzene ring, and the peak at 190.78 ppm belongs to the C atom of $-\text{CHO}$. In the ^{31}P NMR (Figure 2c) of TFMP, there is only one peak located at -18.63 ppm which matches well with the only P atom in the chemical structures of TFMP. As seen from

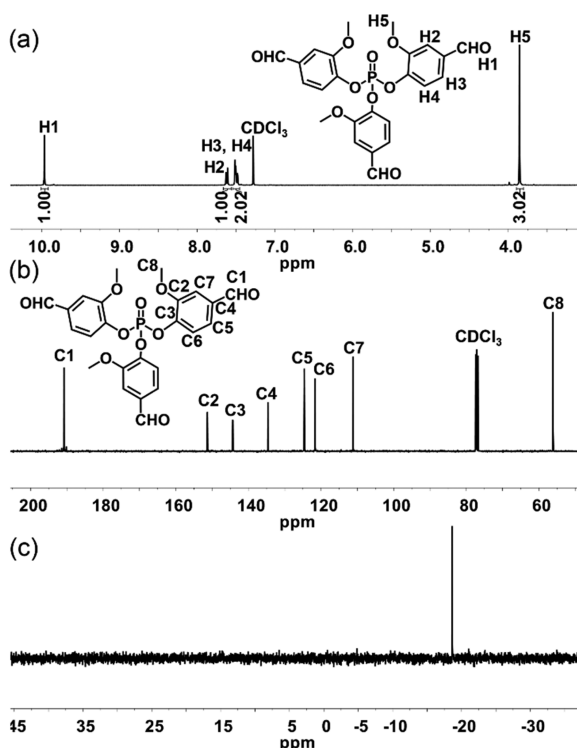


Figure 2. (a) ^1H NMR, (b) ^{13}C NMR, and (c) ^{31}P NMR spectra of TFMP.

the TOF-MS spectrum (Figure S2), the molecular weight of TFMP is $501.0816 \text{ g mol}^{-1}$, which is in accordance with the theoretical molecular weight ($501.09 \text{ g mol}^{-1}$). Besides, the melting point of TFMP determined by DSC (Figure S3) was around 151.2°C . These results are indicative of the successful synthesis of TFMP.

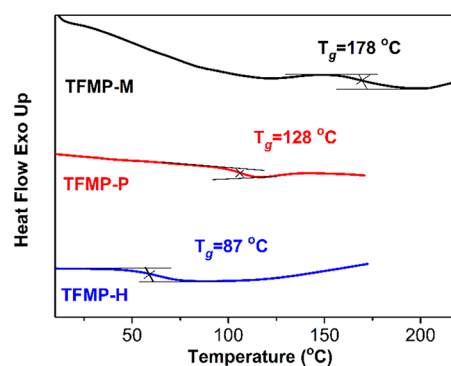


Figure 4. DSC curves of Schiff base CANs.

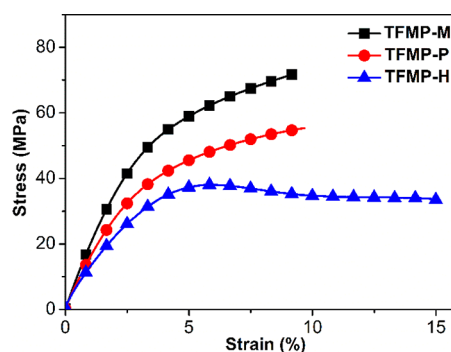


Figure 5. Tensile stress–strain curves of Schiff base CANs.

3.2. Preparation of Schiff Base CANs. Schiff base CANs were obtained by cross-linking TFMP with diamines MDA, PACM, and HMD through Schiff base reaction. To get the optimal curing conditions, real-time FTIR was used to monitor the chemical change during heating. As shown in Figures 3a and 3b, the peak strength for $\text{C}=\text{O}$ of aldehyde groups of TFMP-

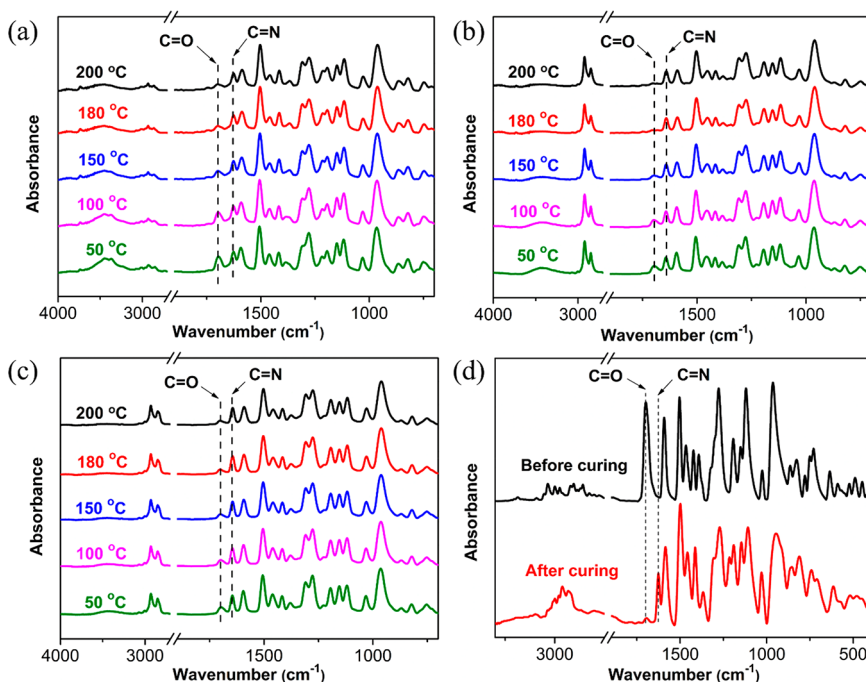


Figure 3. (a–c) Temperature-dependent FTIR measurements of the precured TFMP-M (a), TFMP-P (b), and TFMP-H (c). (d) FTIR spectra of TFMP-M before and after curing.

Table 1. Theoretical Average Molecular Weight between Cross-Link Points (M_c) and the Mechanical Properties of Schiff Base CANs

sample	M_c (g mol^{-1})	tensile strength (MPa)	tensile modulus (MPa)	elongation at break (%)
TFMP-M	744	69.2 ± 3.9	1925 ± 110	8.5 ± 1.1
TFMP-P	762	54.3 ± 1.9	1620 ± 38	9.1 ± 1.3
TFMP-H	621	35.2 ± 3.2	1426 ± 38	15.3 ± 0.9

MPA and TFMP-PACM systems decreased with increasing temperature before 180 °C, while both samples exhibited stable peak strength for C=O of aldehyde groups after heating to 180 °C. This suggests that the curing degree of these two samples should reach to the highest at 180 °C. Thus, they were cured at 180 °C. Although the TFMP-HMD system could reach the highest curing degree before 100 °C reflected from the stable peak strength for C=O of aldehyde group after heating to 100 °C, it was also cured at 180 °C to keep the same curing condition with others.

To investigate the curing degree of the systems, FTIR spectra of TFMP-M after curing are exhibited in Figure 3d. Compared with the FTIR spectra of TFMP, the peaks for C=O at around 1700 cm^{-1} became extremely weak, and the strong peaks for C=N from the Schiff base reaction between C=O and amines at around 1630 cm^{-1} appeared. A similar structure can be seen in TFMP-P and TFMP-H (Figure S4). Besides, the gel contents of three Schiff base CANs are all above 96% (Figure S5). These all indicate that the Schiff base CANs were cured well under the former mentioned conditions.

3.3. Thermal and Mechanical Properties of Schiff Base CANs. Figures 4 and 5 show DSC curves and tensile stress–strain curves of Schiff base CANs, respectively. The data of mechanical properties are listed in Table 1. The cured samples



Figure 7. Reprocess of Schiff base CANs at different times at 180 °C under 15 MPa pressure.

showed superior mechanical properties with tensile strength of 35.2–69.2 MPa, modulus of 1426–1925 MPa, elongation at break of 8.5–15.3%, and high T_g of 87–178 °C. The TFMP-M showed the highest strength, modulus, and T_g and lowest elongation at break, and the TFMP-H showed the lowest strength, modulus, and T_g and highest elongation at break. In general, the strength, modulus, and T_g of the cross-linked polymers increase with the increase of their cross-linked density and chain segment's rigidity. The theoretical average molecular weight between cross-link points (M_c) (Table 1) was calculated using eq 1 to estimate the cross-link density of the cross-linked Schiff base.

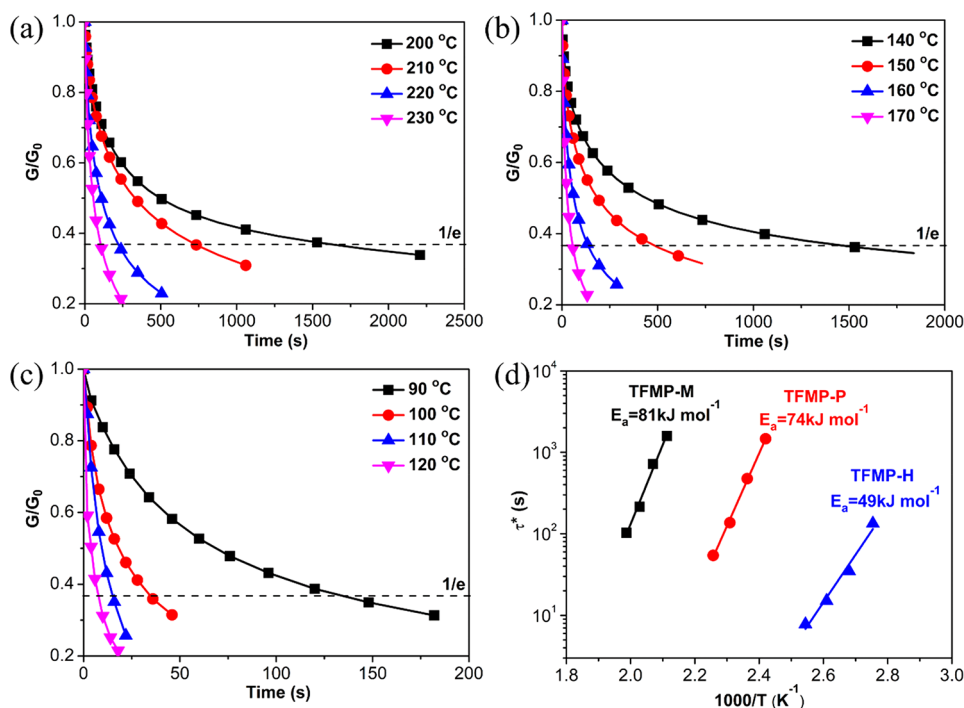


Figure 6. (a–c) Normalized stress relaxation curves of (a) TFMP-M, (b) TFMP-P, and (c) TFMP-H at different temperatures. (d) Arrhenius analysis of the characteristic relaxation time τ^* versus $1000/T$ for the Schiff base CANs.

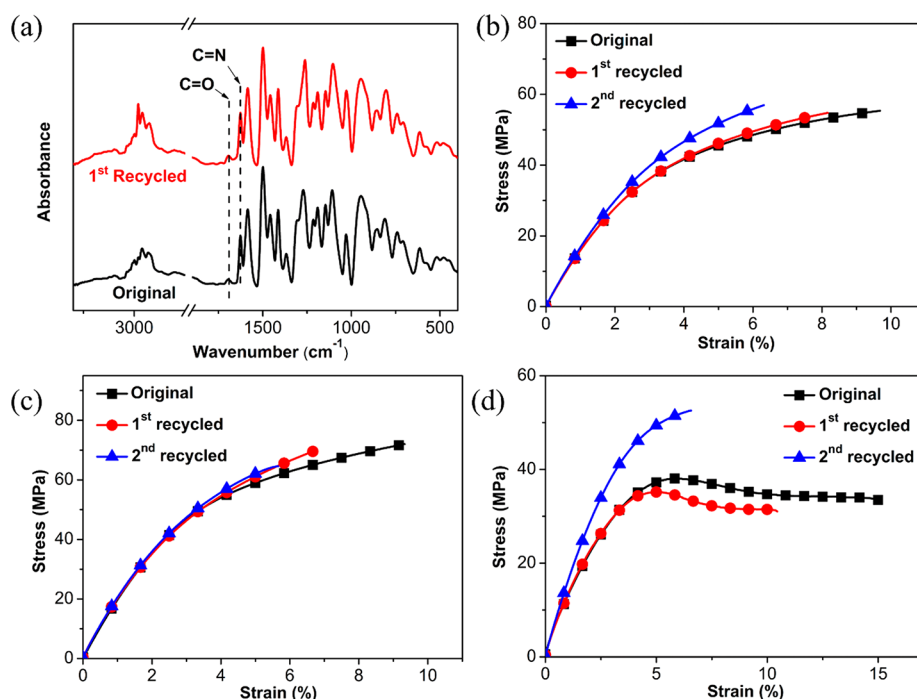


Figure 8. (a) FTIR spectra of original and first recycled TFMP-M. (b–d) Representative tensile stress–strain curves of (b) TFMP-M, (c) TFMP-P, and (d) TFMP-H through two generations of recycling.

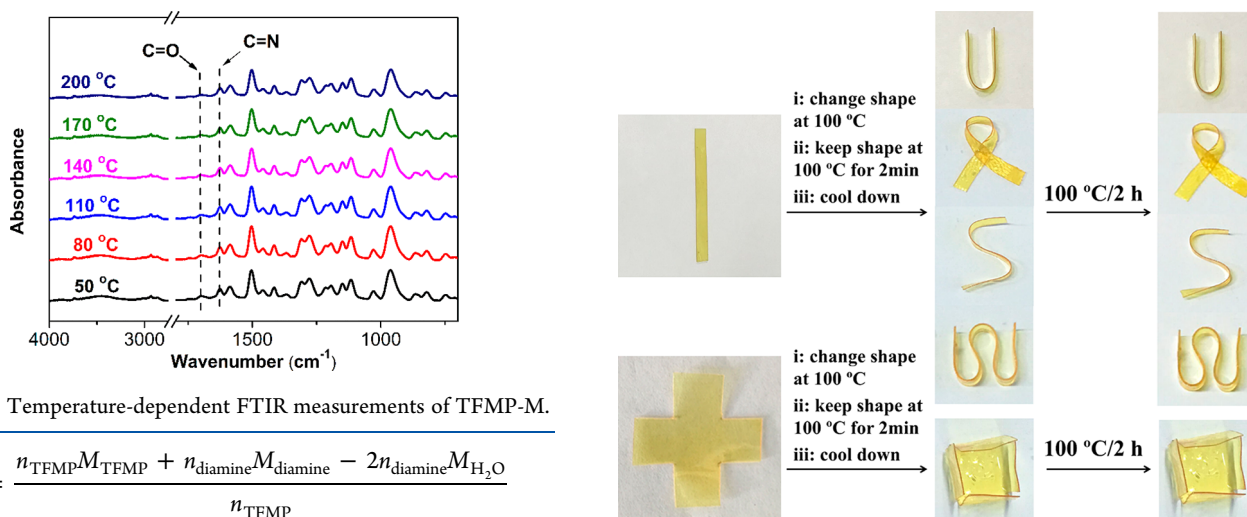


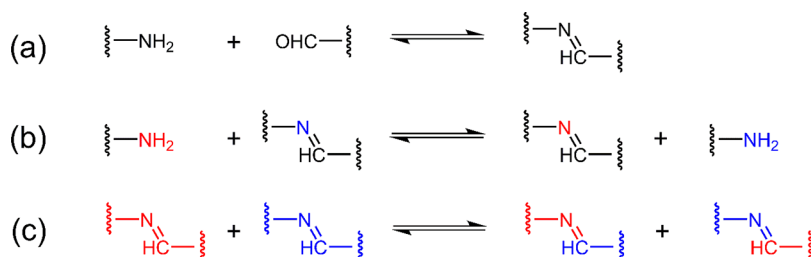
Figure 9. Temperature-dependent FTIR measurements of TFMP-M.

$$M_c = \frac{n_{\text{TFMP}}M_{\text{TFMP}} + n_{\text{diamine}}M_{\text{diamine}} - 2n_{\text{diamine}}M_{\text{H}_2\text{O}}}{n_{\text{TFMP}}} \quad (1)$$

where n and M represent the molarity and molar mass of the corresponding component and the released water. TFMP-H exhibited the lowest M_c of 621 g mol^{−1} and relatively high gel content of 97%, which suggests that its cross-link density should

be the highest, while the TFMP-H owned the flexible long aliphatic chain which might dominate the properties, corre-

Scheme 3. Supposed Reactions during the Reprocess: (a) Imine Formation and Hydrolysis; (b) Transimination; (c) Imine Metathesis



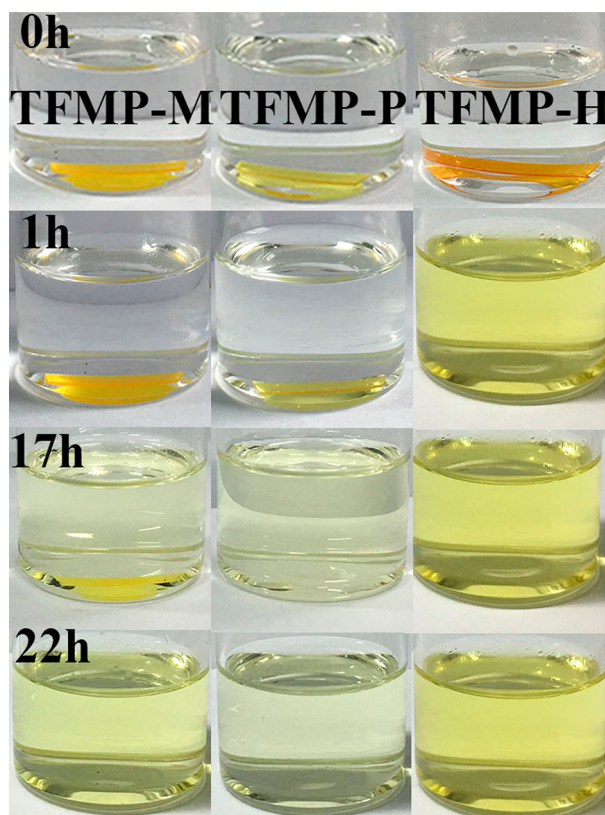


Figure 11. Degradation process of Schiff base CANs at different times in a mixed solution of 1 M HCl aqueous solution and THF (v:v, 2:8).

sponding to its lowest strength, modulus, and T_g and highest elongation at break. TFMP-M possessed a more rigid structure than the TFMP-P owing to the higher rigidity of MDA than PACM⁴⁹ and lower M_c of 744 g mol⁻¹ and higher gel content of 99% than TFMP-P with M_c of 762 g mol⁻¹ and gel content of 96%, leading to the higher cross-link density of TFMP-M than that of TFMP-P; as a result, TFMP-M exhibited the highest strength, modulus, and T_g and lowest elongation at break.

3.4. Malleability of Schiff Base CANs. For the vitrimers or CANs, the malleability is very important to their reprocessability, weldability, reparability, and shape changeability. Thus, the malleability of the Schiff base CANs was investigated by the thermal stress relaxation examining the reduction of stress in response to an external applied strain to the samples.⁹ Figures 6a, 6b, and 6c depict the time- and temperature-dependent

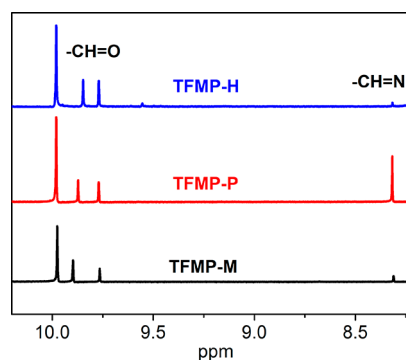


Figure 12. ¹H NMR spectra of Schiff base CANs after degradation.

relaxation modulus of TFMP-M, TFMP-P, and TFMP-H, respectively. All three Schiff base CANs are in accordance with the law of temperature rise and stress relaxation time reduction which can be attributed to the increase exchange rate of Schiff base with increasing temperature; the relaxation times at different temperatures for different samples also illustrate that relaxation ability follows the order of TFMP-H > TFMP-P > TFMP-M. The activation energy (E_a) of the bond exchange reaction was calculated via the Arrhenius equation:⁵⁰

$$\tau^*(T) = \tau_0 \exp(E_a/RT) \quad (2)$$

where the relaxation time τ^* was determined via modulus relaxation to $1/e$. τ_0 is the characteristic relaxation time at infinite temperature, T is the experimental temperature, and R is the universal gas constant. Figure 6d shows the Arrhenius analysis of the characteristic relaxation time τ^* versus $1000/T$ for the Schiff base CANs. The calculated E_a s of TFMP-M, TFMP-P, and TFMP-H were 81, 74, and 48 kJ mol⁻¹, respectively, which was located in the E_a range 33.5–129 kJ mol⁻¹ for polyimines calculated by Zhang's group.²⁷ This result is also in agreement with the relaxation time trend. TFMP-H exhibited the rapidest bond exchange rate due to its best segmental mobility from the flexible structure, and TFMP-M showed the lowest bond exchange rate due to its high content of rigid benzene rings hindering the segmental motion and the relatively stable Schiff base structure from the conjugation with two benzene rings.

3.5. Reprocessing Recyclability. After the malleability of the Schiff base CANs was confirmed, their recyclabilities were explored. As shown in Figure 7, pieces of Schiff base CANs films could be recovered into complete films in 10 min; TFMP-H

Table 2. Summary of the Degradation Time of Schiff Base CANs under Different Conditions

code	Schiff base CANs	temp (°C)	THF (mL)	HCl(aq) (mL)				0.1 M NaOH(aq) (mL)	degradation time
				1.5 M	1 M	0.5 M	0 M		
1	TEMP-M	20	8		2				21 h 15 min
2	TEMP-P	20	8		2				16 h 43 min
3	TEMP-H	20	8		2				48 min
4	TEMP-P	60	8		2				41 min
5	TEMP-P	20	9		1				19 h 36 min
6	TEMP-P	20	7		3				16 h 34 min
7	TEMP-P	20	8	2					16 h 49 min
8	TEMP-P	20	8			2			16 h 58 min
9	TEMP-P	20	8				2		no change after 7 days
10	TEMP-P	20	10						no change after 7 days
11	TEMP-P	20	8					2	no change after 7 days

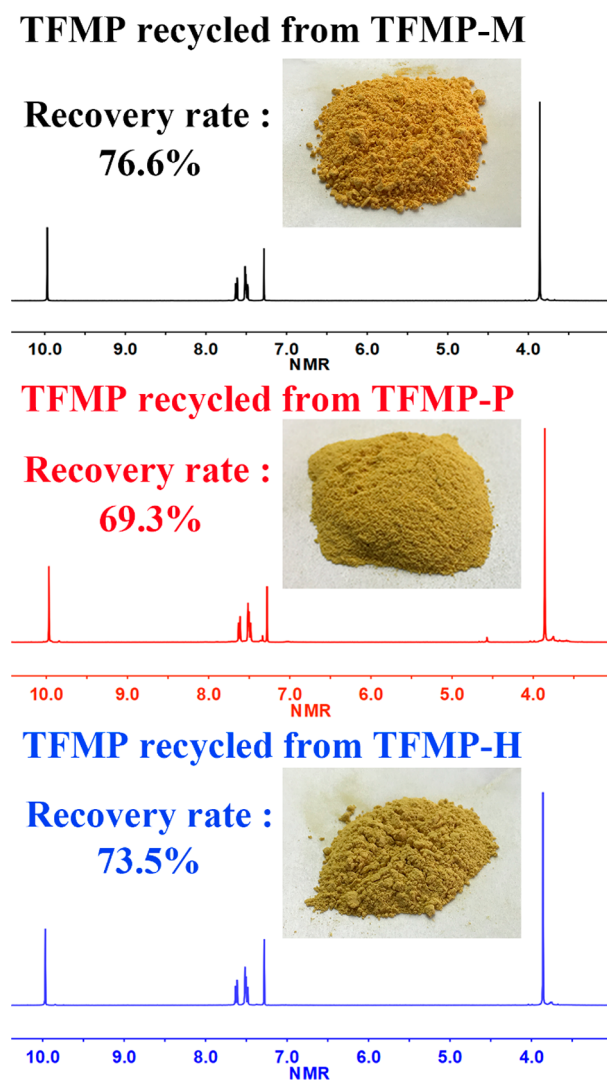


Figure 13. Appearances, recovery rates, and ^1H NMR spectra of TFMP recycled from Schiff base CANs.

required the shortest reprocess time (2 min) while the TFMP-M required the longest one (10 min), which is in agreement with the results of relaxation experiments. At relatively low temperature, pieces of Schiff base CANs could also be reprocessed to complete films with relatively long time, as shown in Figure S6. The characteristic peaks in the FTIR spectra of TFMP-M (Figure 8a), TFMP-P (Figure S7), and TFMP-H (Figure S8) before and after recycling were almost the same, indicating that the chemical structures of the polyimines were preserved well during reprocessing. Figures 8b to 7d show the representative stress-strain curves of the original, first recycled, and second recycled TFMP-M, TFMP-P, and TFMP-H. After recycling through two generations, there is no obvious decrease of mechanical properties, while the moduli increased and elongation at break decreased. This might be due to the side reactions (such as reaction between amine and Schiff base,⁵¹ self-cross-linking of Schiff base,⁵² oxidation, etc.) at high temperatures which would increase the cross-linking density and rigidity of CANs, corresponding to the increased moduli and elongation at break.

To obtain information about the recycling mechanism, a temperature-dependent FTIR measurement was performed for the TFMP-M (Figure 9), TFMP-P (Figure S9), and TFMP-H

(Figure S10). The FTIR spectra of Schiff base CANs at different temperatures from 50 to 200 °C showed no difference especially for the characteristic peaks of $\text{C}=\text{O}$ at around 1700 cm^{-1} and $\text{C}=\text{N}$ at around 1630 cm^{-1} , which suggests that the content of the $\text{C}=\text{N}$ and un-cross-linked $-\text{CHO}$ kept stable. As reported, when $-\text{CHO}$ contacts with $-\text{NH}_2$, they will react to form $\text{C}=\text{N}$ and release water. For TFMP-M, TFMP-P, and TFMP-H, there are un-cross-linked $-\text{CHO}$ and $-\text{NH}_2$ in the cross-linked networks, they should have chance to react with each other to form $\text{C}=\text{N}$ and release water. The unchanged $\text{C}=\text{N}$ and $-\text{CHO}$ content suggests that the same content of $\text{C}=\text{N}$ formed was hydrolyzed by the released water, and the formation and hydrolysis of $\text{C}=\text{N}$ reached equilibrium, as shown in Scheme 3a. The un-cross-linked $-\text{NH}_2$ could also react with $\text{C}=\text{N}$ by amine-imine exchange reaction (transamination, Scheme 3b).⁵³ Imine metathesis (Scheme 3c) should also occur during the reprocess and could be catalyzed by the un-cross-linked primary amine groups.³⁶

3.6. Arbitrary Steady-State Shape Changeability.

Conventional thermosets generally possess the ability to change shapes when heated to their T_g s, while these shapes can only be temporarily fixed because the cross-link network topographies are not changed. When the resins are heated again to temperature range of glass transition, they will be recovered. The Schiff base CANs could undergo arbitrary steady-state or permanent shape change above the T_g s. Take TFMP-H as an example, the programmed shapes kept stable when they were heated at 100 °C for 2 h (Figure 10). This steady-state shape changeability was due to the plasticity via thermally induced network topography rearrangement⁵⁴ contributed by the exchangeable Schiff base structure. For the conventional thermosets, permanent shapes were often obtained by using molds during curing process, machining after curing, or the recently developed 3D printing technology, etc. For these Schiff base CANs, stable or permanent shapes could be easily gotten and adjusted after curing with the help of thermal treatment. This property of Schiff base CANs could increase manufacturing capabilities and flexibility of thermosetting materials.⁵⁰

3.7. Degradation Recyclability and Monomer Recovery.

The cross-linked polyimines can also be hydrolyzed under mild acidic conditions and recovered to the original monomer TFMP and diamines. As shown in Figure 11, Figure S12, and Table 2, TFMP-H degraded and completely dissolved in a mixed solution of 1 M HCl aqueous solution and THF (v: v, 2:8) in only 48 min at 20 °C, TFMP-P and TFMP-M required 16 h 43 min and 21 h 15 min, respectively. Increasing the temperature to 60 °C, the degradation time of TFMP-P was shortened to 41 min. The degradation time shortened when the volume ratio of 1 M HCl aqueous solution and THF changed from 1:9 to 2:8 and 3:7. Kept at the same volume ratio of HCl aqueous solution with THF of 2:8, the different concentrations of HCl aqueous solution (1.5, 1, and 0.5M) almost did not affect the degradation time of TFMP-P at 20 °C (around 16 h 50 min). After degradation, most of $\text{CH}=\text{N}$ were recovered to $\text{HC}=\text{O}$ for TFMP-M, TFMP-P, and TFMP-H as can be seen from Figure 12. So the degradation of the cross-linked Schiff base CANs follows the mechanism of cleaving the Schiff base bonding followed by releasing small molecules which could be dissolved in the mixed solutions. After degradation in the acidic solutions, the original monomer TFMP with high purity could be easily recovered in a simple way and the recovery rate was as high as 76.6% for TFMP-M, 69.3% for TFMP-P, and 73.5% for TFMP-H, as shown in Figure 13. In particular, phosphate structures are

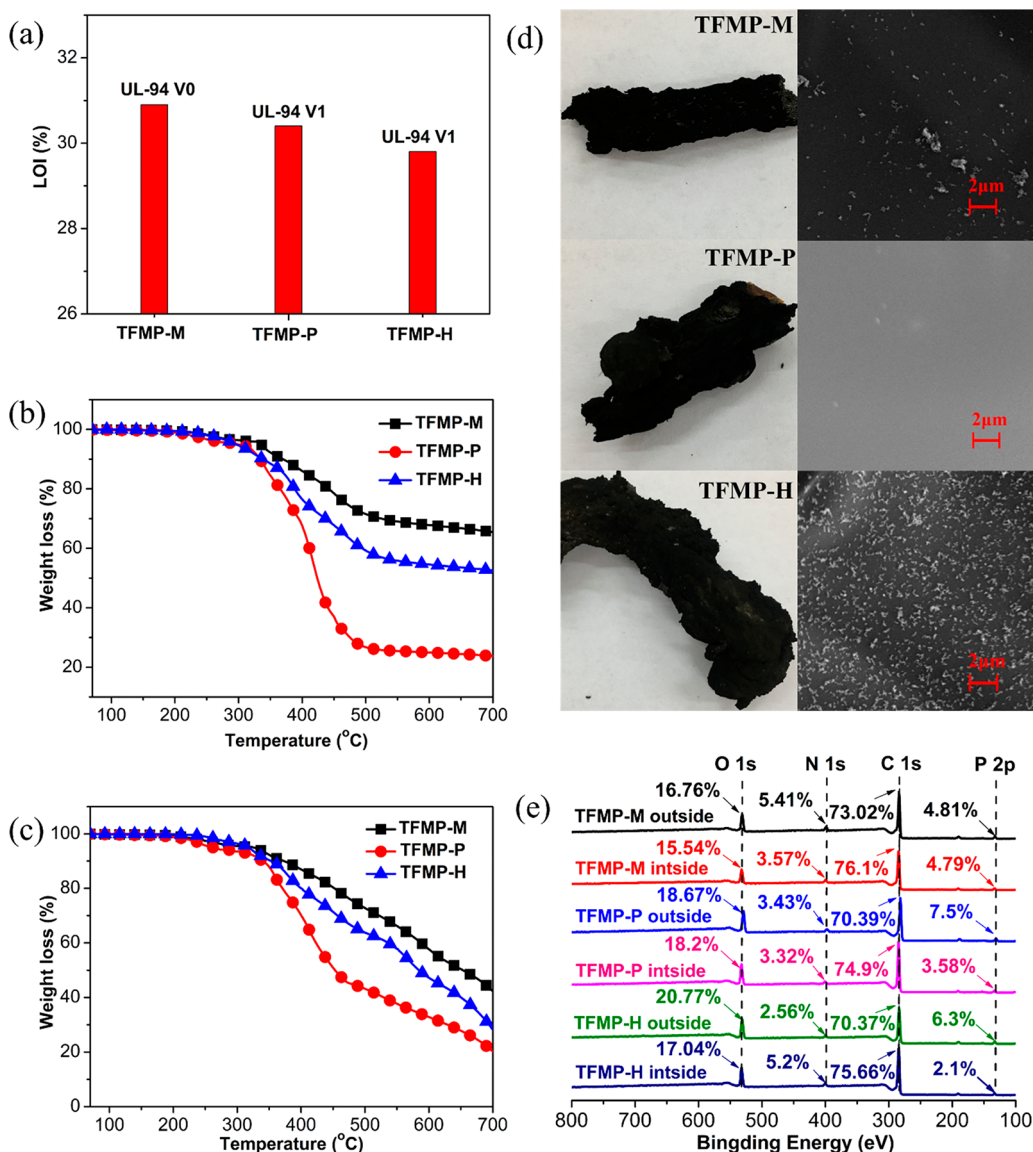


Figure 14. (a) LOI and UL-94 rating. (b) TGA curves under a nitrogen atmosphere. (c) TGA curves under an air atmosphere. (d) Digital photos and SEM photographs of the residues after LOI test. (e) XPS spectra of the surface part and inside part of the char residues.

Table 3. TGA Data and the Phosphorus (P) and Nitrogen (N) Content before Burning of the Schiff Base CANs

sample	$T_{d5\%}$ (°C)		$T_{d30\%}$ (°C)		R_{700} (%)		before burning	
	air	N ₂	air	N ₂	air	N ₂	P (%)	N (%)
TFMP-M	335	326	522	525	42.3	65.5	4.17	5.65
TFMP-P	299	263	401	395	20.2	23.7	4.04	5.51
TFMP-H	296	318	457	437	28.2	52.6	4.99	6.77

also degradable. When TFMP was placed in a mixed solution of 1 M aqueous solution and acetone- d_6 (v:v, 2:8), about 1/18 of TFMP determined by the ^{31}P NMR spectra in Figure S11 was degraded after 24 h. Thus, the TFMP could not be fully recovered is mainly due to the degradation of phosphate structure in TFMP. Besides, TFMP-P showed the relatively lower recovery rate compared with TFMP-M and TFMP-H because more $\text{CH}=\text{N}$ were maintained for TFMP-P after degradation which can be seen from its higher proton peak in the ^1H NMR spectrum of the TFMP-P after degradation than other two samples (Figure 12). Besides the excellent recyclability, materials also require essential stability for utilization. Thus, the

chemical stability of the Schiff base CANs under other conditions were also investigated. Take TFMP-P as an example; it was stable in either neutral or alkaline conditions which can be reflected from the stable appearances after immersing in THF, THF/ H_2O mixture solvent, and THF/ NaOH aqueous mixture solution at 20 °C for 7 days (Table 2).

3.8. Fire Resistance. As shown in Figure 14a, all the Schiff base CANs exhibited excellent flame retardancy; TFMP-M showed the best with the highest fire resistant rate of UL-94 V0 during vertical burning test and limit oxygen index (LOI) value of 30.9%, and TFMP-P and TFMP-H both reached the UL-94 V1 rate and showed LOI values of 30.4% and 29.8%,

respectively. The flame retardancy of Schiff base CANs could be comparable with other conventional fire-resistant polymers which are mainly through the addition of flame retardants. The better flame retardancy of TFMP-M relative to TFMP-P and TFMP-H was attributed to its higher content of benzene rings beneficial for carbonization, which can be seen from its higher carbon residue at 700 °C by TGA examination. To disclose the flame-retardant mechanism of these three Schiff base CANs, the thermal degradation was investigated by TGA first (Figure 14b,c and Table 3). The initial degradation temperatures (degradation temperature for 5% weight loss, $T_{d5\%}$) of the three Schiff base CANs are relatively low (close to 200 °C), which is ascribed to the easily thermally degradable O=P–O bond.⁵ While all the samples exhibited high degradation temperature for 30% weight loss ($T_{d30\%}$) and char yield at 700 °C (R_{700}) higher than the commonly used bisphenol A epoxy resins.^{5,55,56} The thermal degradation of the O=P–O bond at lower temperature led to the formation of phosphorus-rich surface char layer which acted as a superior thermal insulating layer and prevented heat reaching the inside of the samples,^{46,57} corresponding to the high $T_{d30\%}$ and R_{700} . And the high C=N content in the samples also contributed to the high char yield.⁵⁸ TFMP-M displayed much higher carbonization ability than TFMP-P and TFMP-H as can be seen from its much higher $T_{d30\%}$ and R_{700} , resulting in its better flame retardancy than TFMP-P and TFMP-H. Flammability characteristics of polymeric materials can also be reflected by the structure of char residue after combustion. As can be seen from the digital photos and SEM photographs of the char residues after LOI test (Figure 14d), TFMP-M, TFMP-P, and TFMP-H all formed great intumescent char residues, and the char layers of TFMP-M, TFMP-P, and TFMP-H are extremely dense. To further identify the chemical components of the char residues, the surface part and the inside part of the char residues were examined by XPS (Figure 14e). Obviously, there are much more phosphorus and oxygen and less carbon in the surface part of the char residues than in the inside part of the char residues for all the three Schiff base CANs. This result is similar to our previous reported phosphorus-containing epoxy resins.⁵ Before burning, the calculated nitrogen content is 5.65% for TFMP-M, 5.51% for TFMP-P, and 6.77% for TFMP-H (Table 3), and all decreased during combustion as can be seen from XPS spectrum (Figure 14e), which indicates that the nitrogen-containing structures were also degraded and played the gas source during the combustion. The Schiff base structure of TFMP-M located between two benzene rings is more stable than that of TFMP-P and TFMP-H; as a result, more nitrogen was locked on the surface char layer of TFMP-M with a nitrogen content of 5.41%. It can be concluded that the excellent flame retardancy of the samples is mainly from the great and extremely dense intumescent char formation ability during the combustion which is contributed from the high content of phosphorus-containing structure and Schiff base structure. The phosphorus-containing structure degraded at low temperature (reflected from the low $T_{d5\%}$ by TGA), transferred to the surface, captured the radicals containing oxygen, and remained in the surface layer of the residues (reflected from the higher phosphorus and oxygen content of the surface char residues than that of the inside char residues by XPS); Schiff base structure provided the gas source and also had contributed to the high char yield.

4. CONCLUSIONS

Our efforts toward addressing recycle and flame retardancy of thermosets from bioresources have the potential to foster the

development of using renewable biomasses and their platform chemicals to build environmentally benign and advanced materials. For the first time, high-performance, flame-retardant, and highly malleable thermoset Schiff base CANs that were not only reprocessable but also readily recyclable to monomers were prepared from a novel Schiff base precursor based on lignin derivative vanillin. While for other reported malleable thermosets, high performance, readily reprocessability, and monomers recovery were difficult to be achieved simultaneously. The Schiff base CANs displayed excellent performance with high T_g , strength, and modulus and could be readily recycled by two methods of reprocessing within 2–10 min and degradation to original monomer TFMP with high recovery rate of around 70% under mild conditions and exhibited excellent flame retardancy as high as UL-94 V0 rate during vertical burning. Besides, an arbitrary change of steady-state shape was achieved above T_g in virtue of the dynamic covalent Schiff base structure, which could increase manufacturing capabilities and flexibility of thermosetting materials. Through this work, we also found that some side reactions (reaction between amine and Schiff base,⁵¹ self-cross-linking of Schiff base,⁵² oxidation, etc.) may occur during the remolding process, resulting in brittleness increase of the materials, which is unfavorable for the use of materials. In addition, rapid relaxation is beneficial to the recovery of materials but also limits the use of materials at high temperatures. These two issues need to be urgently addressed in the future.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.macromol.8b01601.

Nonisothermal DSC curves and TOF-mass spectrum of TFMP; photos of precured, postcured, and recycled Schiff base CANs; FTIR spectra, gel content, temperature-dependent FTIR and degradation process of Schiff base CANs; ³¹P NMR spectra of TFMP in acid solution (PDF)

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Notes

The authors declare no competing financial interest.

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